

Yuliang Peng

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EDUCATION

University of Maryland, College Park

- Bachelor of Science in Computer Science (Dean's List)
- Master of Science in Machine Learning
- Relevant Courses: Machine Learning, Computer Vision, Robotics, Deep Learning, Data Structures and Algorithms

Graduated: May 2024

Expected: May 2027

WORK EXPERIENCE / RESEARCH EXPERIENCE

University of Maryland, College Park *Research Assistant*

Maryland, USA

Nov 2025 - Present

- Developed GPU-accelerated spatiotemporal embeddings for high-rate **event camera** streams, enabling scalable ML inference via **CUDA**-based parallel processing.
- Conducted data-driven analysis of aviation network resilience using **network-level modeling** on Delta system outage data.
- Applied **NLP** and sentiment analysis to TripAdvisor travel reviews to extract passenger perception dimensions

ModeSens *Software Engineer Full-Time*

Remote, USA

Jun 2025 - Sep 2025

- Built a **distributed** e-commerce crawler with asynchronous pipelines and **serverless** triggers, ingesting product data from **30+** merchant catalogs..
- Enhanced an **NLP** pipeline to normalize user search queries against merchant catalogs, improving search precision by **13%**.
- Implemented an **image-based** product matching workflow that compares user-uploaded images against internal product catalogs to support **visual search** and recommendations.

Bank of China, USA *Software Engineer Full-Time*

Los Angeles, USA

July 2024 - May 2025

- Led bug bash and reliability efforts for internal enterprise systems, resolving **70+** high-impact issues and reducing productivity-blocking tickets by **10%**.
- Implemented a secure client authorization framework using cryptographic signatures and the Chain of Responsibility pattern, adopted by **15+** internal teams.
- Built **AWS**-based data pipelines (EC2, S3, RDS) for reliable real-time ingestion and storage of financial transaction data.

Air China *Software Engineer Intern*

Sichuan, China

May 2023 - August 2023

- Developed **Unix** shell scripts to automate system backups across Linux environments, reducing manual workload by **40%**.
- Revamped the secure login system with mobile QR code scanning, streamlining user identification. (adopted in 2024)

PROJECTS

Tabit – AI-Powered Browser Tab Manager | Google Chrome Extension Hackathon | [Github Code](#)

- Built an **AI**-powered Chrome extension (React, Vite, Manifest V3) to automatically cluster and organize browser tabs.
- Implemented similarity-based **tab grouping** with hierarchical clustering and Web Worker parallel processing to manage **100+** tabs in real time.
- Designed a **session memory system** to persist and restore tab groups across browsing sessions.

FaceID-Style Verification System | LFW Dataset | [Github Code](#)

- Built an end-to-end **face verification system** on the **LFW dataset** using similarity-based embedding matching.
- Designed a **reproducible ML pipeline** for data ingestion and pair generation with ROC-based threshold calibration.
- Packaged the inference workflow using **CLI interface** and benchmarked latency under concurrent workloads.

Image-Based Product Matching Tool | Machine Learning | [Github Code](#)

- Built an **image similarity system** using perceptual hashing and SSIM to match product images.
- Developed a Python-based image extraction pipeline with Selenium, reducing manual matching workload by **80%** and improving recommendation accuracy by **25%**.

SKILLS

Programming Languages: C++, CUDA, Python, C, Java, Dart, JavaScript, Unix Shell, SQL

Tools / Frameworks: AWS, Docker, Redis, NumPy, Pandas, Flask, Spring Boot, Vue.js, Flutter, Git, Sentry